

Coordination Cages



Dynamic Stereochemistry of M_8Pd_6 Supramolecular Cages Based on Metal-Center Lability for Differential Chiral Induction, Resolution, and Recognition

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Abstract: A series of isostructural supramolecular cages with a rhombic dodecahedron shape have been assembled with distinct metal-coordination lability (M_8Pd_6 -MOC-16, $M = Ru^{2+}$, Fe^{2+} , Ni^{2+} , Zn^{2+}). The chirality transfer between metal centers generally imposes homochirality on individual cages to enable solvent-dependent spontaneous resolution of Δ_8/Λ_8 - M_8Pd_6 enantiomers; however, their distinguishable stereochemical dynamics manifests differential chiral phenomena governed by the cage stability following the order $Ru_8Pd_6 > Ni_8Pd_6 > Fe_8Pd_6 > Zn_8Pd_6$. The highly labile Zn centers endow the Zn_8Pd_6 cage with conformational flexibility and deformation, enabling intrigue chiral- Δ_8/Λ_8 - Zn_8Pd_6 to meso- $\Delta_4\Lambda_4$ - Zn_8Pd_6 transition induced by anions. The cage stabilization effect differs from inert Ru^{2+} , metastable Fe^{2+}/Ni^{2+} , and labile Zn^{2+} , resulting in different chiral-guest induction. Strikingly, solvent-mediated host-guest interactions have been revealed for Δ_8/Λ_8 -($Ru/Ni/Fe$) $_8Pd_6$ cages to discriminate the chiral recognition of the guests with opposite chirality. These results demonstrate a versatile procedure to control the stereochemistry of metal-organic cages based on the dynamic metal centers, thus providing guidance to maneuver cage chirality at a supramolecular level by virtue of the solvent, anion, and guest to benefit practical applications.

Introduction

Chiral phenomena play an important role in living systems for many essential biological processes, such as molecular recognition, information transmission and chemical transformation.^[1] Therefore, the exploration of external stimuli and internal factors that are able to affect chiral

assembly and interactions in artificial systems has triggered great interest in understanding the chirality origin and the recognition laws of nature, opening up new applications in enantiomers separation, chiral sensing, and asymmetric transformation^[1c,2] by intelligent control and manipulation of chirality.^[3] In imitation of biological microenvironments, the synthetic metal-organic cages (MOCs) have been developed as the artificial paradigms owing to their facilely designable structures featuring in various exo- and endo-cavities for host-guest interactions.^[1c,4] Many chiral-MOCs have been successfully obtained either through integration of enantiomeric ligand or metalloligand precursors,^[1a,c,2a,3d,5] or through induction of chiral host-guest interactions.^[6] However, the examples for protein mimicking with characteristics of guest-binding pockets remain limited, especially the chiral-MOCs possessing multiple binding-sites for chiral regulations at a supramolecular level through multiple guest interactions.

As the proteins are intrinsically dynamic, metal cations of distinct labile nature can provide a valuable source to investigate dynamic properties of the assembled cages for protein-mimicking, allowing twisted configurations around metal centers to increase diversity of cage structures^[7] with stereochemical merits.^[7a,8] The six-coordinated first-row transition metals, such as Fe^{2+} , Ni^{2+} and Zn^{2+} , have been widely applied in cage assembly, focusing on the shape and size control of the cage containers;^[3i,9] however, their significance in cage dynamics and functions should not be neglected.^[3k,10] When it comes to a large cage composed of multiple metal centers, the tiny change on individual metal center may be amplified or synergized, leading to macroscopically visible disparities. For example, the labile Zn^{2+} ions show versatile and flexible coordination geometries, thereof becoming one of the most essential components in a number of enzymes with irreplaceable role in metabolism.^[11] Recently, Nitschke and co-workers demonstrated the dynamic divergence brought about by the alteration of the metal centers from Fe^{2+} to Zn^{2+} .^[6b,12] Nonetheless, systematic research on the implication of metal lability for the cage stereochemical dynamics, especially for multi-guest binding regulation, remains elusive. Unveiling the variances brought about by metal ions in cage assembly may facilitate more precise cage structure design and mastery over guest binding modes, therefore catering to diverse application scenarios.

We previously reported stepwise assembly of the rhombic dodecahedral Δ_8/Λ_8 -($Ru/Fe/Zn$) $_8Pd_6$ (MOC-16

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series),^[6c,13–16] which afforded enantiopure cages by pre-resolution of the metalloligand precursors^[14] or in situ chiral induction in the secondary assembly,^[6c] respectively, owing to the distinct metal lability. The twelve open-pockets and one large inner cavity make these cages good candidates for multi-guest binding, showing promising applications in enantioseparation^[6c,14] and asymmetric catalysis.^[4c,15] Herein we systematically demonstrated the stereochemistry of this type of cages inherent in the coordination lability of metal stereocenters (designated as M_8Pd_6 -MOC-16) including inert Ru^{2+} ($4d^6$), metastable Fe^{2+} ($3d^6$) and Ni^{2+} ($3d^8$), and labile Zn^{2+} ($3d^{10}$) for comparison (Figure 1). A fruitful supramolecular chiral chemistry has been manifested by this cage series in terms of chiral induction, resolution and recognition, which are manipulatable by anions, guests and solvents based on the dynamic nature of metal centers, thus providing a full stereochemical scenario of this cage family and a helpful guidance to the rational design and application of chiral-MOCs containers.

Results and Discussion

Four rhombic dodecahedron-shaped M_8Pd_6 cages of different metal centers were prepared according to our previously reported metalloligand stepwise assembly strategy^[6c,13,16]

(Figures 2 and S1–S17, see experimental details in Supporting Information). The high resolution electrospray ionization mass spectrometry (HR ESI-MS) measurements in acetonitrile (CH_3CN) solution confirm that all cages display a similar series of successively charged species corresponding to $[M_8Pd_6^{28+} - xH^+ + yBF_4^-]^{(28-x-y)+}$ (Figures 2c, S10, S12, S14 and S16). Each species involves a distribution of same charged entities associating different numbers of counter anions and protons, which is verified by a comparison of the experimental and simulated isotopic patterns. The 1H NMR spectra of Ru_8Pd_6 and Fe_8Pd_6 show a neat set of proton resonances in CD_3CN – D_2O (1/4, v/v) with signals moved up- or down-field diversely in comparison with those of metalloligands (Figures 2b, S13 and S15), indicating that these metal centers are diamagnetic in low-spin d^6 configuration. In contrast, the paramagnetic Ni^{2+} ions in Ni_8Pd_6 cage cause severely shifted and broadened resonances (Figures 2b and S9); while Zn_8Pd_6 cage gives a 1H NMR pattern containing other minor peaks, implying existence of a small quantity of Zn-cage in different conformations (Figures 2b and S11, vide infra).

The single-crystals of all M_8Pd_6 cages suitable for X-ray diffraction analysis were grown from the slow diffusion of methanol (CH_3OH) into the CH_3CN solution containing in situ assembled cages and KPF_6 in two weeks (except for Zn_8Pd_6 that was obtained without KPF_6). Since the pre-

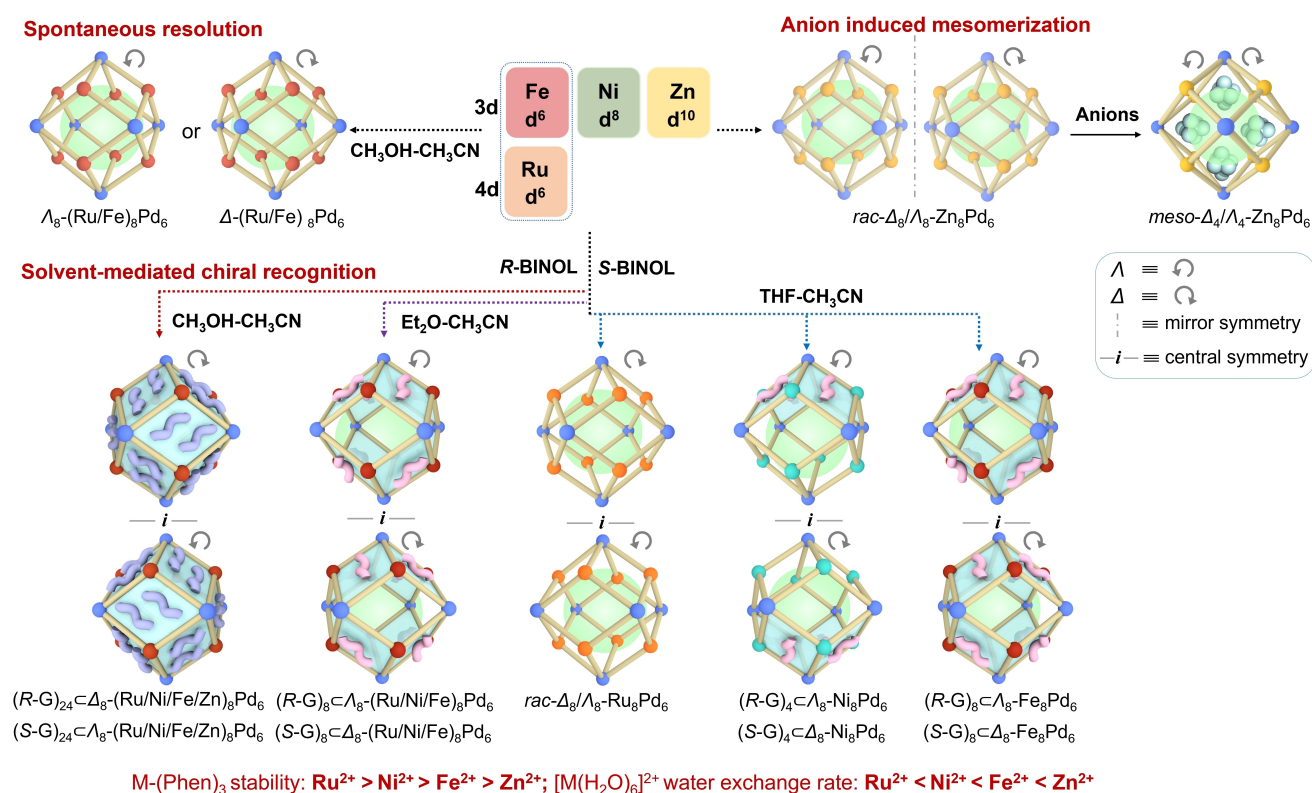


Figure 1. Representative chiral regulations of M_8Pd_6 cages based on metal dynamics. The Ru_8Pd_6 and Fe_8Pd_6 cages containing metal centers of d^6 electronic configuration undergo spontaneous resolution in CH_3OH – CH_3CN solution to form enantiomorphic crystals, while giving racemic crystals in Et_2O – CH_3CN and THF – H_2O mixtures. The highly flexible Zn_8Pd_6 cage of $3d^{10}$ metal centers is able to achieve anion induced mesomerization. The host–guest chiral recognition is governed by solvophobic effects, with opposite chiral guest binding behaviors, subtly influenced by the metal lability. In the diagram, cages are depicted by rhombic dodecahedra and guests are depicted by wiggly bars.

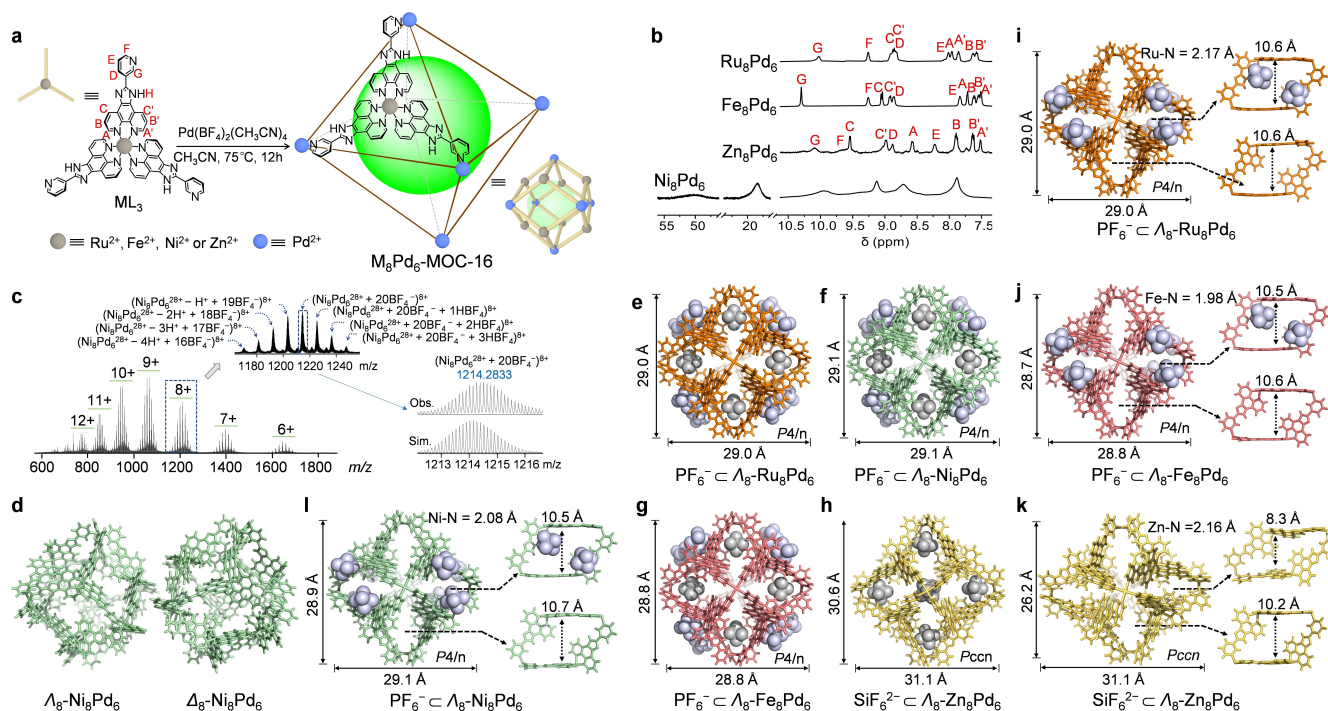


Figure 2. (a) Metalloligand stepwise assembly of M_8Pd_6 -MOC-16 cages. (b) Partial 1H NMR spectra of M_8Pd_6 cages (400 MHz, CD_3CN/D_2O , 1/4, v/v, 298 K). (c) HR ESI-MS spectra recorded in a positive mode of the Ni_8Pd_6 cage in CH_3CN . The inserts show the distribution of +8 charged species with associated anions, and a comparison of the simulated and measured isotopic signals for one species. (d) X-ray single-crystal structures of a pair of enantiomers of Λ_8 - and Δ_8 - Ni_8Pd_6 cages that co-crystallize in a racemic crystal. (e–h) Top view of the X-ray single-crystal structures of anion- Λ_8 - M_8Pd_6 showing the encapsulated anions and overall cage sizes. The pocket binding anions are presented by blue white balls and interior-vertex binding anions by grey balls. Other anions outside the cage are omitted for clarity. (i–l) Side view of the X-ray single-crystal structures of anion- Λ_8 - M_8Pd_6 showing the pocket binding anions on the equatorial plane of the cage (blue white balls). Other anions are omitted for clarity. The pocket widths with and without anions are depicted.

assembled tris-bidentate ML_3 metalloligands feature in a classical fan-like configuration, the secondary Pd^{2+} -driven assembly should start from a pair of opposite chiral precursors. As we previously observed, chiral transmission between metal centers takes place to result in formation of homochiral Δ_8 - or Λ_8 - M_8Pd_6 cages in equal amount in solution,^[6c,13,16] however, all four cages crystallize in the form of racemic *rac*- Δ_8/Λ_8 - M_8Pd_6 compounds with both enantiomers coexisting in the same crystals^[17] (Figures 2d, S17, and Tables S4, S7, S9 and S11). The averaged M–N bond lengths follow an order of $4d^8-Ru^{2+}$ (2.17 Å) > $3d^{10}-Zn^{2+}$ (2.16 Å) > $3d^8-Ni^{2+}$ (2.08 Å) > $3d^6-Fe-N$ (1.98 Å), giving rise to overall cage sizes of Ru_8Pd_6 (29.0×29.0×29.0 Å), Zn_8Pd_6 (31.1×30.6×26.2 Å), Ni_8Pd_6 (29.1×29.1×28.9 Å) and Fe_8Pd_6 (28.8×28.8×28.7 Å) (Figures 2e–l, based on opposite Pd...Pd distances). The Zn_8Pd_6 cage shape obviously deviates from regular polyhedron due to conformational freedom imposed by the labile Zn^{2+} ions, showing compressed pockets on the equatorial plane of the cage. Under each Pd-vertex in the inner cavity, there is one anion encapsulated; moreover, there are 8 anions captured in the 4 equatorial pockets to form anion- M_8Pd_6 complexes.

We noticed that the solvent has a significant impact on the crystal growth habit.^[16] Without additional KPF₆, the as-prepared cages with BF_4^- as counter anions can crystallize in either racemic compounds or enantiopure compounds,^[18]

and part of anions are displaced by SiF_6^{2-} due to HBF_4 induced glass dissolution. When diffusing ethyl ether (Et_2O) into the CH_3CN solution, or tetrahydrofuran (THF) into the H_2O solution of as-prepared cages, the racemic *rac*- Δ_8/Λ_8 - M_8Pd_6 compounds are always obtained.^[6c,19] Interestingly, when the diffusion solvent was changed to CH_3OH , the chiral spontaneous resolution process was observed for Ru_8Pd_6 and Fe_8Pd_6 cages, resulting in the formation of enantiopure crystals^[17] (Figures S18–20 and Tables S12–13). The homochiral Δ_8 -(Ru/Fe)₈ Pd_6 or Λ_8 -(Ru/Fe)₈ Pd_6 enantiomers crystallize in separate single-crystals in $P4_22$ chiral space groups, leading to more loose and porous packing of the cages in contrast to the dense packing in racemic crystals (Figure S20). On the contrary, the Zn_8Pd_6 cage crystallizes in the form of racemate in $Pccn$ centrosymmetric space group under the same conditions^[17] (Figure S21 and Table S4). We speculate that this is due to flexibility of the Zn-cage which presents dynamic conformation variations in solution, thus preventing accumulation of homochiral cages during crystallization.^[20] Unfortunately, the single crystal of Ni_8Pd_6 cage suitable for X-ray diffraction analysis was not obtained under similar conditions. Nevertheless, these results suggest that the CH_3OH solvent facilitates spontaneous resolution of the relatively rigid homochiral cages.^[21] Such solvent-mediated spontaneous resolution process is rare, since only about 5–10 % of racemic molecules are able

to spontaneously resolved,^[18b] which are mainly from small molecules and seldom happens for racemic MOCs through so-called symmetry breaking process.^[21–22] As the process of crystal nucleation and growth involves complex thermodynamic and kinetic factors,^[18a,23] predicting whether spontaneous resolution crystallization can occur for a racemate remains challenging.

As the counter anions electrostatically interact with the cage in interior, exterior and pocket sites, their impact on the cage conformation was investigated for four M_8Pd_6 cages (Figure S22). By adding PF_6^- in co-crystallization up to 160 equivalents (equiv.), we observed a significant conformation change in Zn_8Pd_6 , namely, the *rac*- Δ_8/Λ_8 - Zn_8Pd_6 cages were able to be converted into an achiral *meso*- $\Delta_4\Lambda_4$ - Zn_8Pd_6 (Figures 3, S23–25 and Table S4).^[17] More anion-cage interactions are generated by the presence of additional PF_6^- anions, which may motivate the fluxional rearrangement of metalloligands around labile Zn-centers in solution. Therefore, a conformational interconversion between Δ - and Λ - ZnL_3 may occur. The single-crystal structural analysis unveils that *meso*- $\Delta_4\Lambda_4$ - Zn_8Pd_6 comprises 4 Δ - ZnL_3 and 4 Λ - ZnL_3 metal centers with opposite chiral configurations. The homochiral cage pockets of C_2 -symmetry in Δ_8/Λ_8 - Zn_8Pd_6 have been transformed into mirror-symmetric pockets composed of two opposite chiral metal centers (Figure S23). Two closely adjacent ligands are enforced in this mesomeric cage, thereby favoring formation of O–H...N and N–H...O hydrogen bonds to curb a water molecule by two imidazole moieties (Figure 3c).

Compared to the broad 1H NMR profile of *rac*- Δ_8/Λ_8 - Zn_8Pd_6 , the dissolved *meso*- $\Delta_4\Lambda_4$ - Zn_8Pd_6 - PF_6 crystals exhibits complex but assignable signal patterns in CD_3CN (Figures 3d and S25). The proton resonances are split into

two sets, one belonging to the ligand protons outside the cage while the other belonging to those inside the cage. **This is consistent with the low symmetry of the cage as observed in crystal structure, thus generating chemically inequivalent environments of these protons.** It is noteworthy that a new exchangeable proton peak appears at 14.3 ppm, which is attributed to the bound water molecule^[24] (Figure S25); moreover, the imidazole-NH signal is down-field shifted due to hydrogen binding. The 1H -DOSY measurement confirms that *meso*- $\Delta_4\Lambda_4$ - Zn_8Pd_6 - PF_6 can exist as a sole species in CD_3CN solution (Figure S25), which keeps stable in the CD_3CN solution at room temperature for 1.5 months.

Similar anion-induce mesomerization of the Δ_8/Λ_8 - Zn_8Pd_6 cages also happens by using excessive SbF_6^- and ClO_4^- anions, affording *meso*- $\Delta_4\Lambda_4$ - Zn_8Pd_6 - SbF_6 and *meso*- $\Delta_4\Lambda_4$ - Zn_8Pd_6 - ClO_4 crystals when adding 90 and 30 equiv. SbF_6^- and ClO_4^- , respectively^[17] (Figures 3, S26–29, and Tables S1–2, S5–6). The distinct color and morphology changes are discerned for this mesomerization process (Figures 3b, S26 and S28), which facilitates visual identification of the *meso*- $\Delta_4\Lambda_4$ -cage crystals without the aid of the X-ray diffractometer or NMR spectrometer.^[25] Furthermore, the crystallization process is also dependent on the nature of the anions. For example, the tetrahedral ClO_4^- promotes faster crystal growth to decrease the time from approximately two weeks of PF_6^- to four days. In contrast, the attempt to induce chirality transformation of $(Ru/Ni/Fe)_8Pd_6$ cages failed,^[17] indicating that the inert Ru-stereocenter and the metastable Ni/Fe-stereocenter integrated in the cage framework are robust enough to resist anion induction (Figures S30–32, and Tables S7–11).

We further explored the potential solution transformation process from Δ_8/Λ_8 - M_8Pd_6 to *meso*- $\Delta_4\Lambda_4$ - M_8Pd_6 (Fig-

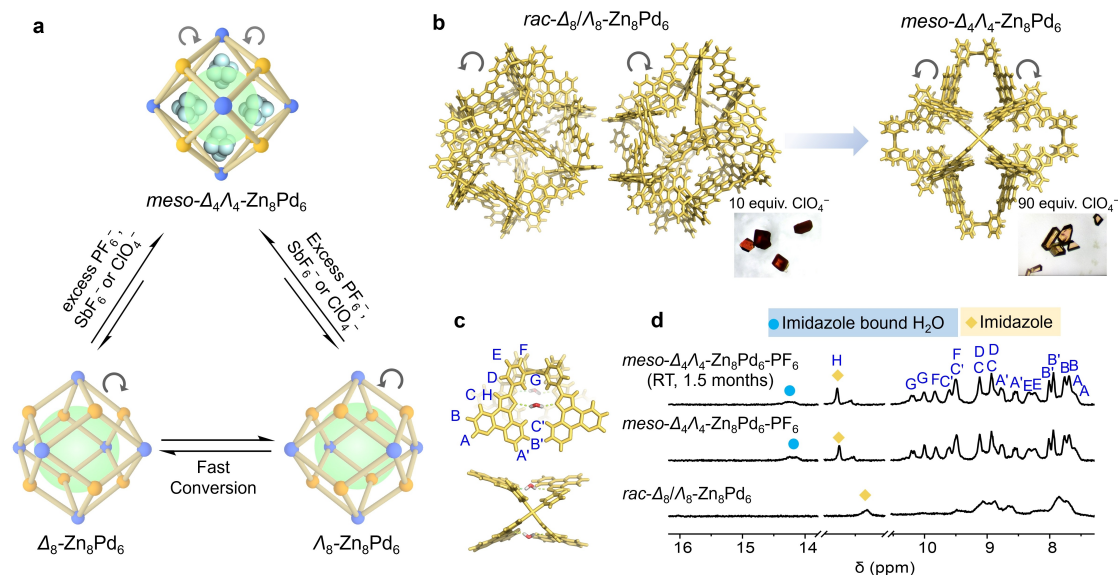


Figure 3. (a) Schematic illustration of the transformation of Zn_8Pd_6 stereoisomers with the aid of excess anions. (b) X-ray crystal structures of *rac*- Δ_8/Λ_8 - Zn_8Pd_6 and *meso*- $\Delta_4\Lambda_4$ - Zn_8Pd_6 . The inserts show photographs of crystals grown in the presence of different amounts of ClO_4^- . (c) The imidazole bound water observed in the crystal structure of *meso*- $\Delta_4\Lambda_4$ - Zn_8Pd_6 . (d) 1H NMR spectra of *rac*- Δ_8/Λ_8 - Zn_8Pd_6 , *meso*- $\Delta_4\Lambda_4$ - Zn_8Pd_6 - PF_6 crystals, and *meso*- $\Delta_4\Lambda_4$ - Zn_8Pd_6 - PF_6 crystals stored at room temperature in CD_3CN for 1.5 month (400 MHz, CD_3CN , 298 K). Yellow solid rhombus: imidazole-NH signal; blue solid circles: imidazole bound water-OH signal.

ure S33–44), which can be monitored by the new imidazole bound water-OH signal characteristic of *meso*- Δ_4A_4 -cage. To confirm this water binding behavior by two neighboring imidazoles via hydrogen bonds in *meso*- Δ_4A_4 - Zn_8Pd_6 , we carried out 1H - 1H NOESY and HR ESI-MS spectral characterization (Figures S25c, S27c and S29c,d). It is clear that the water- H^w resonances show through-space correlation with H^C and H^G . The mass spectrum also verified the presence of water bound *meso*-cage species in CH_3CN solution. As expected, it turns out that the PF_6^- induction effect is not effective in solution. Even a large amount of PF_6^- anions cannot change the 1H NMR pattern of Δ_8/Δ_8 - Zn_8Pd_6 (Figure S33). **This means the mesomerization energy barrier retards PF_6^- induction, which needs assistance from a crystallization process to move the mesomerism equilibrium.** For comparison, the SbF_6^- and ClO_4^- anions can efficiently induce mesomerization process, leading to structural conversion as observed by appearance of the water-OH signal when adding 12 or 28 equiv. SbF_6^- and ClO_4^- , respectively (Figures S34–35). **Nonetheless, Δ_8/Δ_8 - Zn_8Pd_6 and *meso*- Δ_4A_4 - Zn_8Pd_6 cages seem to coexist in the solution.** Similar to the observations in crystallization, the Δ_8/Δ_8 -(Ru/Ni/Fe) $_8Pd_6$ cages could not be induced by anions to undergo structural conversion in solution (Figures S36–44), indicating that the metastable Fe^{2+}/Ni^{2+} -stereocenters are solidified by a cage stabilization effect.^[6c]

The chiral host–guest interactions have tested by co-crystallization of *rac*- Δ_8/Δ_8 - M_8Pd_6 cages with (*R*)-(+)-1,1'-bi-2-naphthol (*R*-BINOL) and (*S*)-(–)-1,1'-bi-2-naphthol (*S*-BINOL) enantiomeric guests. **The results reveal remarkable solvophobic effects on the chiral recognition behaviors,**^[4d,16] giving rise to three different guest encapsulation modes (G:H=24: 1, 8: 1 and 4: 1) with opposite chiral recognition,^[6c] depending on the difference in solvents and metal centers^[17] (Figures 1 and S45–49). In a CH_3OH - CH_3CN solvent system, the *R*-BINOL guests induce crystallization of the (*R*-BINOL) $_{24}\Delta_8$ - M_8Pd_6 clathrates, while the *S*-BINOL guests select Δ_8 -cages to form the (*S*-BINOL) $_{24}\Delta_8$ - M_8Pd_6 clathrates. Totally 24 BINOLs are dually packed in the 12 chiral pockets of the cage, where the guests are captured through the O–H...N hydrogen bonds between BINOL and imidazole-N, as well as the cage-guest and guest-guest π - π stacking interactions (Figures 4a, S50–55 and Table S14–16). All Ru/Ni/Fe/Zn-cages are competent for this kind of host–guest chiral selection.^[6c,16–17,26]

As the solvent conditions are able to influence crystallization process and host–guest interactions,^[3a,16,27] we also tried co-crystallization in an Et_2O - CH_3CN solvent system (Figures 1, S56–59 and Tables S17–18). Strikingly, the same chiral guest inducers present reverse host–guest recognition behaviors,^[6c,16] i.e., the *R*-BINOL guests choose to co-crystallize with Δ_8 - M_8Pd_6 , while the *S*-BINOL guests select Δ_8 - M_8Pd_6 .^[17] The captured guests are decreased from 24 to 8, and only the top and bottom pockets of the cage are occupied by a single BINOL molecule to lead to crystallization of the (*S*-BINOL) $_8\Delta_8$ - M_8Pd_6 and (*R*-BINOL) $_8\Delta_8$ - M_8Pd_6 clathrates (Figure 4b). The middle pockets are empty and slightly compressed. Interestingly, two planar naphthalene moieties of one BINOL are

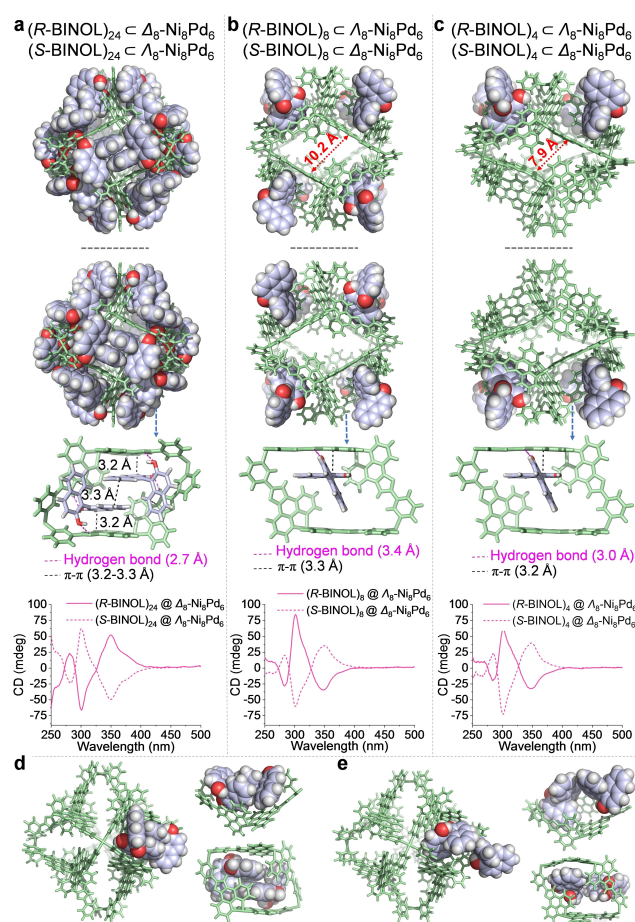


Figure 4. X-ray single-crystal structures, CD spectra (CH_3CN , 298 K), and simulations for the chiral host–guest recognition of Δ_8/Δ_8 - Ni_8Pd_6 cages. (a) *R*-guest $\subset \Delta_8$ -cage vs. *S*-guest $\subset \Delta_8$ -cage recognition with G:H=24:1 in CH_3OH - CH_3CN solvent. (b) *R*-guest $\subset \Delta_8$ -cage vs. *S*-guest $\subset \Delta_8$ -cage recognition with G:H=8:1 in Et_2O - CH_3CN solvent. (c) *R*-guest $\subset \Delta_8$ -cage vs. *S*-guest $\subset \Delta_8$ -cage recognition with G:H=4:1 in THF - CH_3CN solvent. (d) Geometry optimization with two *S*-BINOL molecules in a pocket of the Δ_8 - Ni_8Pd_6 cage, showing mutual fit-in enclathration. (e) Geometry optimization with two *R*-BINOL molecules in a pocket of the Δ_8 - Ni_8Pd_6 cage, showing the poor fit.

embedded into the pockets of two neighboring cages to form O–H...N hydrogen bonds and cage-guest π - π stacking interactions, thus connecting two same chiral cages in the crystal lattice (Figures S56–59). Despite numerous attempts, guest-induced crystallization of *rac*- Δ_8/Δ_8 - Zn_8Pd_6 in the Et_2O - CH_3CN system was unsuccessful, suggesting that the conformational fluctuation of Zn-cage disfavors the enclathration of one guest in its pockets to sustain its homochirality during crystallization process.

The chiral host–guest interactions become even more complicated in a THF - CH_3OH solvent system, which are also influenced by the metal centers (Figures 1, S60–65 and Tables S19–21). Surprisingly, the *rac*- Δ_8/Δ_8 - Ru_8Pd_6 cages crystallize alone without binding guest molecules in this solvent condition^[17] (Figures S60–61), while the *rac*- Δ_8/Δ_8 - Zn_8Pd_6 cages could not crystallize as observed in the Et_2O - CH_3CN media. The *rac*- Δ_8/Δ_8 -(Ni/Fe) $_8Pd_6$ cages

display a similar chiral guest selection as in $\text{Et}_2\text{O}-\text{CH}_3\text{CN}$, that is, *R*-BINOL selectively co-crystallizes with $\Delta_8-(\text{Fe}/\text{Ni})_8\text{Pd}_6$ while *S*-BINOL chooses $\Delta_8-(\text{Fe}/\text{Ni})_8\text{Pd}_6$.^[17] However, the guest: host ratios (G: H) are different in these two cases. Only four top pockets of the Ni-cage are occupied by the single BINOL guests (Figures 4c and S62–63). The empty middle pockets are compressed (7.6–7.9 Å) while the empty bottom pockets keep normal conformation. In contrast, the Fe-cages capture 8 BINOL guests in each of its top and bottom pockets, leaving the middle pockets free (Figures S64–65). These results imply a more dynamic host–guest interactions in $\text{THF}-\text{CH}_3\text{CN}$ solvent system, probably due to the superior solubility of BINOL molecules in THF, which weakens the enclathration ability. **Under such circumstance, host–guest binding behaviors become more sensitive to the conformational freedom of the cages inherent in the lability of metal centers.**

To further understand the unusual host–guest chiral recognition with opposite selection by the single-guest binding and double-guest binding in one pocket, we performed the Materials Studio simulations for the geometry optimization based on the single-crystal model (Figures 4d–e). The results suggest that two *S*-BINOL molecules could be well accommodated in a pocket of Δ - Ni_8Pd_6 cage with complementary fit-in selection; while two *R*-BINOL molecules are mismatched in one pocket of Δ - Ni_8Pd_6 cage, squeezing one guest out of pocket to release crowding packing. Therefore, the double-guest binding mode favors a *R*-guest Δ -cage or *S*-guest Δ -cage recognition behavior, while the single-guest binding mode prefers a *R*-guest Δ -cage or *S*-guest Δ -cage recognition behavior, which have been confirmed by the circular dichroism (CD) spectral measurements of the dissolved crystals in CH_3CN to show a pair of mirror signals exactly correlated with the Δ - or Λ -metal centers (Figures 4a–c). The protic CH_3OH may have more prominent solvophobic effect to enforce BINOL enclathration,^[4d] resulting in fully filled G_{24}CH as thermodynamic species in solution. On the contrary, the aprotic Et_2O and THF manifest poorer solvophobic effect, facilitating faster guest exchange in solution to reduce guest binding affinity.^[4d] Nevertheless, the low guest-loading G_8CH and G_4CH species tend to aggregate in a homochiral way by sharing the BINOL guests, thus inducing co-crystallization of the enantiopure *R*-guest Δ -cage or *S*-guest Δ -cage compounds. Such solvent-mediate opposite chiral recognition and induction in crystallization process is rare,^[3a] which may offer a new route to generate diastereoselective host–guest systems for chiral purification and catalysis on a supramolecular level.

The enantio-differentiation recognition behaviors of $\Delta_8/\Lambda_8-\text{Fe}_8\text{Pd}_6$ cages for the *R/S*-BINOL guests have been checked in CD_3CN solution as representative. The ^1H NMR spectral profile of *R*-BINOL $\Delta_8-\text{Fe}_8\text{Pd}_6$ is resembling with its *S*-BINOL $\Delta_8-\text{Fe}_8\text{Pd}_6$ enantiomer, but distinct from its diastereomeric *R*-BINOL $\Lambda_8-\text{Fe}_8\text{Pd}_6$ (Figures S66 and S71). This means the chiral cage can selectively recognize *R*- and *S*-BINOL guests to exhibit distinguishable host–guest dynamics,^[6c] which also reflects in the chemical shifts, namely, *R*-BINOL is more shielded by the Δ -cage than by

the Λ -cage, and vice versa for the *S*-BINOL. A larger up-field shift of guest signals is observed in $\text{CD}_3\text{CN}-\text{D}_2\text{O}$ mixture (1/2, v/v) than in CD_3CN solution, indicating a stronger host–guest affinity caused by the hydrophobic effect.^[4d] This chiral recognition has been further testified by the ^1H NMR titration of $\Delta_8/\Lambda_8-\text{Fe}_8\text{Pd}_6$ cages with *R/S*-BINOL guests one-by-one (Figures S67–70 and S72–75), showing proton movements along aliquot addition of guest into host solution with similarity in enantiomeric host–guest complexes but difference between diastereomeric pairs. The ESI-MS spectra revealed a multiply guest binding behavior in solution (Figure S76), displaying a series of satellite peaks corresponding to $(\text{BINOL})_{14-31}\Delta_8\text{Fe}_8\text{Pd}_6$ host–guest complexes centered around the guest-binding number of 24.

From the above results and discussion, we can see that the stereochemistry of this series of M_8Pd_6 cages shows differential dynamics in solution, which are closely related to the lability of the metal centers and the fluxion of the coordination spheres. The stepwise assembly of the cages from *rac*- ML_3 precursors follows a chirality transfer and amplification rule to give rise to homochiral $\Delta_8/\Lambda_8-\text{M}_8\text{Pd}_6$ structures;^[1c,6c,28] however, their chiral stability in solution strongly depends on the metal lability.^[6c,16] The CD spectral monitoring of the enantioexcess of the fresh $\Delta_8/\Lambda_8-\text{M}_8\text{Pd}_6$ crystals dissolved in CH_3CN indicates a chiral stability order of $\text{Ru}_8\text{Pd}_6 > \text{Ni}_8\text{Pd}_6 > \text{Fe}_8\text{Pd}_6 > \text{Zn}_8\text{Pd}_6$ (Figure 5), which is consistent with the trend in the stability constants of 3:1 metal complex of phenanthroline.^[9a,29] This stability order also coincides with the ligand-exchange rate constants of hexaaqua metal complexes (Figure 1),^[30] signifying the importance of coordination dynamics on these metal centers for solution chiral robustness. The $\Delta_8/\Lambda_8-\text{Zn}_8\text{Pd}_6$ cages obviously exhibit a rapid racemization process, losing the CD intensity to a large extent in 1 day at room temperature. The signals of $\Delta_8/\Lambda_8-\text{Fe}_8\text{Pd}_6$ cages decrease by about 27% under the same conditions, while a greater stabilizing effect

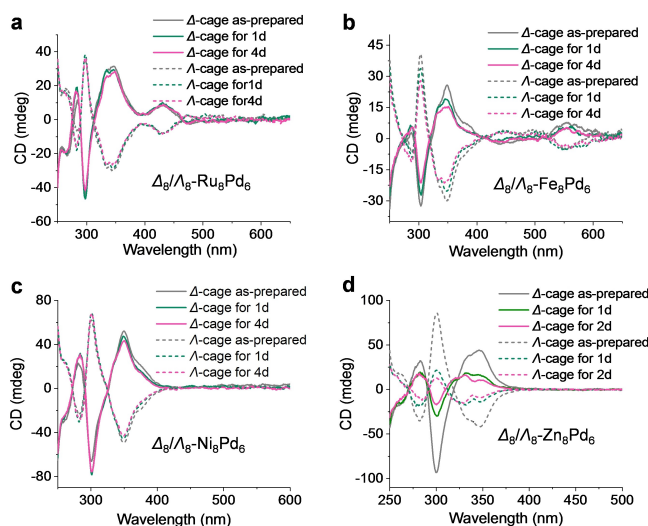


Figure 5. The CD signal changes over time of homochiral $\Delta_8/\Lambda_8-\text{Ru}_8\text{Pd}_6$ (a), $\Delta_8/\Lambda_8-\text{Fe}_8\text{Pd}_6$ (b), $\Delta_8/\Lambda_8-\text{Ni}_8\text{Pd}_6$ (c), and $\Delta_8/\Lambda_8-\text{Zn}_8\text{Pd}_6$ (d) crystals dissolved in CH_3CN solution.

is found for Ni-center,^[9a] showing decrease of the signals by about 10 % in one day. For comparison, the Δ_8/Δ_8 -Ru₈Pd₆ cages retain most CD intensity in 4 days, owing to the inertness of the Ru²⁺-centers. Considering the stereoconfigurational lability of the free (Ni/Fe)L₃ metal complexes that are apt to undergo dissociation and racemization in solution,^[31] the reasonable chiral stability of the Δ_8/Δ_8 -(Ni/Fe)₈Pd₆ cages is attributable to the cage stabilization effect via mechanical coupling among the metal centers to inhibit the conformational changes.^[6c,32] These observations also account for the reason why Δ_8/Δ_8 -Zn₈Pd₆ cages of high dynamic nature could be induced by excessive anions to undergo mesomerization, but Δ_8/Δ_8 -(Ru/Ni/Fe)₈Pd₆ cages did not.

Since the chiral guest binding may alter the equilibrium between dynamic Δ_8/Δ_8 -M₈Pd₆ enantiomers and promote conversion to one of the dominant configurations,^[6a-c,33] we further investigate the in situ chiral induction of the homochiral cages by introducing *R/S*-BINOL into the assembly process from *rac*-ML₃^[6c] (Figures S77–80 and Table S3). 9 equiv. of *R/S*-BINOLs were added as the inducer to the metalloligands solution, and the cages were precipitated from the solution by adding Et₂O. The CD spectra were then recorded after dissolving the resulting cages in CH₃CN. It turned out that such in situ induction strategy is effective for metastable Fe₈Pd₆ and labile Zn₈Pd₆ cages, leading to enantiomeric products with the chiral configuration same as the cages obtained under the CH₃OH–CH₃CN co-crystallization condition (Figures S77–78). This observation suggests that the Pd²⁺-driven secondary assembly of the cages from *rac*-(Fe/Zn)L₃ precursors is governed by the *R/S*-BINOL inducers, forming (*R*-BINOL)₂₄⊂ Δ_8 -(Fe/Zn)₈Pd₆ complexes induced by *R*-BINOL, or, forming (*S*-BINOL)₂₄⊂ Δ_8 -(Fe/Zn)₈Pd₆ complexes induced by *S*-BINOL, as thermodynamically predominate products in solution. The CD measurements also disclose that the in situ induced Δ_8/Δ_8 -Fe₈Pd₆ cages have a better enantioexcess than Δ_8/Δ_8 -Zn₈Pd₆ cages, which implies a competitive equilibrium between rapid racemization and chiral-induction in the solution, resulting in a lower enantiopurity of Δ_8/Δ_8 -Zn₈Pd₆ due to highly labile Zn²⁺-centers. This has been testified by a comparison of the racemization rates between Δ_8 -Fe₈Pd₆ and Δ_8 -Zn₈Pd₆ cages in CH₃CN, confirming that the CD signal of Δ_8 -Zn₈Pd₆ decreases fast in 50 min with the racemization rate constant $k_r = 1.93 \times 10^{-4} \text{ s}^{-1}$, while Δ_8 -Fe₈Pd₆ keeps almost unchanged in 190 min (Figure S81–82). By contrast, the in situ induction of Δ_8/Δ_8 -Ni₈Pd₆ cages is ineffective (Figure S79), owing to a relatively higher stability of Ni²⁺-centers.^[9a,29a,34] The in situ induction completely does not work for Δ_8/Δ_8 -Ru₈Pd₆ (Figure S80), due to strong inertness of Ru²⁺-centers. These experimental results suggest that the in situ guest-induced chiral resolution of M₈Pd₆ cages is sensitive to the lability of the metal centers, offering a potential route for fast and scaleup preparation of dynamic Δ_8/Δ_8 -(Fe/Zn)₈Pd₆ cages.

Conclusion

We have demonstrated versatile chiral regulations of four multi-pocket (Ru/Ni/Fe/Zn)₈Pd₆ cages characteristic of distinct metal center lability. In this way, we have correlated diverse chiral induction, resolution and recognition behaviors with the chiral stability and stereochemical dynamics of the cages in the order of Ru₈Pd₆ > Ni₈Pd₆ > Fe₈Pd₆ > Zn₈Pd₆. The solvents play multiple roles not only to result in the spontaneous resolution of Δ_8/Δ_8 -M₈Pd₆ cages, but also to trigger rare host–guest interactions to lead to diastereoselective recognition for *R*-G⊂*A*-H and *S*-G⊂*A*-H versus *R*-G⊂*A*-H and *S*-G⊂*A*-H enantiomeric pairs. The anions are able to drive a unique *chiral*- Δ_8/Δ_8 -Zn₈Pd₆ to *meso*- Δ_4/Δ_4 -Zn₈Pd₆ transformation based on the highly labile Zn-stereocenters. The chiral guests not only show single-guest and double-guest binding modes to induce co-crystallization of M₈Pd₆ cages in reverse homochirality, but also induce in situ chiral assembly of Δ_8/Δ_8 -M₈Pd₆ cages. The dynamic stereochemistry of the M₈Pd₆ cages described in this work presents a full scenario of chiral manipulation based on the nature of metal centers, providing insight for the design and control of chiral metal-organic cages and useful guidance for their feasible applications in fields such as chiral sensing, enantiomer separation, and asymmetric transformation.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

Keywords: Chiral Induction • Chiral Recognition • Coordination Cage • Host-Guest Chemistry • Spontaneous Resolution

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